

(quoting Airtel's price). In this article I will mainly talk about iPhone 3G, but it can be applied to other devices as well, with a few exceptions.

The iPhone

iPhone is, in a way, probably the world's best smartphone till date. Earlier, it used to be locked and developers were only allowed to build Safari Web applications. (Safari is Apple's Web browser based on Apple's Webkit, which is based on KDE's KHTML engine.) Arggh! Because of this horrible limitation I was not looking at iPhone as a serious opportunity for developers. But Apple has changed for the better. Not only has it opened the iPhone APIs, but has also constantly improved the SDK with thousands of new APIs accompanying each release.

For developers, this is a dream come true. With iPhone OS 3.0 slated for release in mid-2009, there should soon be a 1,000 new APIs for iPhone, which will enable developers to do cool things:

- **In App Purchase:** Enable you to purchase additional content from within the applications. A good example would be games, where someone can purchase an extra level from within the game. This means an extra revenue source for developers.
- **Apple Push Notification Service:** A way by which an application can trigger events, and send messages and alerts even when the application is not running. A good example would be an IM client in which you can log in and close the application, and still receive IM notifications.
- **External Accessory Framework:** Hardware developers can use this API to give their device a great face and powerful ARM CPU. Devices can communicate over iPhone's 30-pin dock connector and wirelessly over Bluetooth. An excellent example was shown by a medical company at the iPhone OS 3.0 pre-release announcement event. The company had connected its blood sugar checking device to iPhone to create a full blown life-saving application for diabetic patients, which offered suggestions on what to eat, how much to exercise and what medicines to take. Useful, isn't it?
- **P2P Device Connectivity:** A great set of APIs that allows multiple iPhone devices to interact with each other over Bluetooth, without pairing. Again, at the iPhone OS 3.0 pre-release event, the developer of the music app, Smule, connected multiple devices running Smule to produce orchestra-like music.
- **Maps (and location APIs):** Developers can build applications that will interact with users based in their locality. They can integrate Google Maps within their applications. Imagine a food application telling you the address for your favourite cuisine that's available close by.

These are just a few of the *new* APIs that will be available in iPhone OS 3.0—many others are already available in the current version of iPhone OS/SDK.

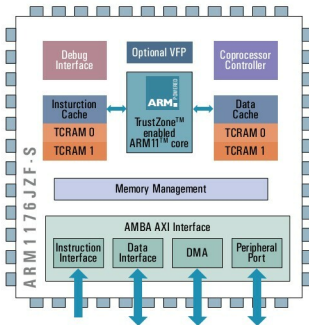


Figure 1: The ARM 1176JZF architecture

iPhone hardware

That's not just all; iPhone is powered by insanely powerful hardware. Let's look at the specs:

- CPU: 620 MHz ARM 1176, under-clocked to 412 MHz
- GPU: PowerVR MBX Lite 3D
- Storage capacity: Flash memory (Original: 4, 8, & 16 GB; 3G: 8 & 16 GB)
- Memory: 128 MB DRAM
- Display: 480×320 px, 3.5 inch (89 mm) colour LCD with an aspect ratio of 3:2, capable of up to 262,144 colours.
- Input: Dock connector, headphone jack, Wi-Fi (802.11b/g), Bluetooth 2.0+EDR
- Camera: 2.0 mega-pixels
- Connectivity: Quad-band GSM 850 900 1800 1900 GPRS/EDGE
- 3G also includes: Tri-band UMTS/HSDPA 850, 1900, 2100, A-GPS
- Accelerometer: Detects the movement in hardware and reacts accordingly.

Let's talk about the CPU to begin with. Yes, it is ARM—don't you guys get sick of seeing it all the time? This is the second time I am talking about ARM in *LFY*; the first instance was for the Nokia 6630 (*Outsmart Your Smart Phone*, published way back in June 2006). In short, ARM is the CPU for all next-generation devices these days, including Nintendo DSi (the game console), Nokia smartphones (6630, N73, N95), iPods, and so on.

The one included in iPhone is even better:

- The ARM1176JZF chip with TrustZone (enables a trusted computing environment for media, apps, network, OS, etc). Yes, not a lot of people like the trusted-environment bit—and the reasoning is quite